

Student: Toheeb Eji
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Instructor: Aleksandra Gajic
Course: ECON 1BB3: Introductory
Macroeconomics

Assignment: Chapter 7

1. The following table gives real GDP figures for an economy for four years.

Fill in the growth rate for each year, and then find the average annual growth rate for the period.

Enter all answers as percentages rounded to one decimal place.

Year	Real GDP	Growth Rate (%)
1999	150 billion	–
2000	153.0 billion	2.0 %
2001	159.1 billion	4.0 %
2002	165.5 billion	4.0 %

The average annual growth rate for this period is %.

2. Why do economic growth rates matter?

- ☐ A. High growth rates coincide with improved living standards.
- ☐ B. When a country sustains high growth rates, life expectancy at birth increases.
- ☐ C. High levels of sustained economic growth reduce infant mortality.
- ☒ D. All of the above.
- ☐ E. A and C only.

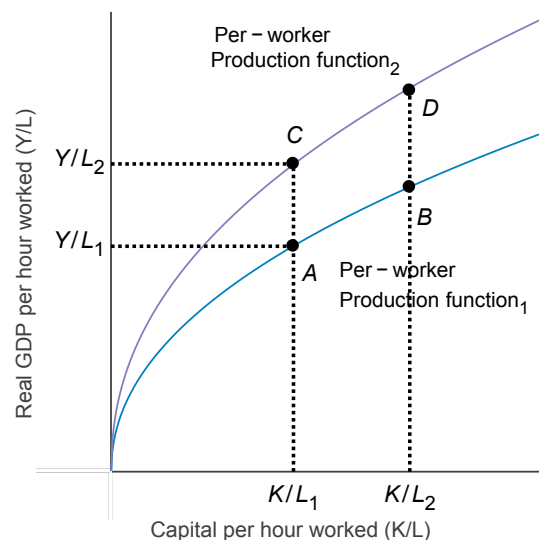
3. Consider the per-worker production function graph.

If there is an increase in capital per hour worked, holding technology constant, then

- ☐ A. there is a movement from *D* to *C*.
- ☒ B. there is a movement from *A* to *B*.
- ☐ C. there is a movement from *A* to *C*.
- ☐ D. there is a movement from *B* to *D*.

If there is an increase in technology, holding constant the quantity of capital per hour worked, then

- ☐ A. there is a movement from *D* to *B*.
- ☒ B. there is a movement from *A* to *C*.
- ☐ C. there is a movement from *C* to *D*.
- ☐ D. there is a movement from *A* to *B*.



4. The *new growth theory* states that

- ☐ A. firms will add to an economy's stock of physical capital leading to increases in real GDP but at a decreasing rate.
- ☐ B. knowledge capital is subject to decreasing returns at the level of the whole economy but increasing returns at the firm level.
- ☒ C. firms will add to an economy's stock of knowledge capital by engaging in research and development or by contributing to technological change.
- ☐ D. growth in labor productivity is a significant factor in bringing about long-run growth in real GDP per capita.

The new growth theory differs from the growth theory developed by Robert Solow, since

- ☐ A. the new growth theory says that physical capital is nonrival and nonexcludable and the Solow growth theory states that knowledge capital is nonrival and nonexcludable and is a key determinant of economic growth.
- ☐ B. the new growth theory focuses on technological change and the quantity of capital available to workers whereas the Solow growth
- ☐ C. the Solow growth theory says that the rate of technological change is influenced by how individuals and firms respond to economic incentives, whereas the new growth theory leaves technological change unexplained.
- ☒ D. the Solow growth theory focuses on technological change and the quantity of capital available to workers whereas the new growth theory states that accumulation of *knowledge capital* is a key determinant of economic growth.

5. Firms are likely to underinvest in research and development, which slows the accumulation of knowledge capital, slowing economic growth, because

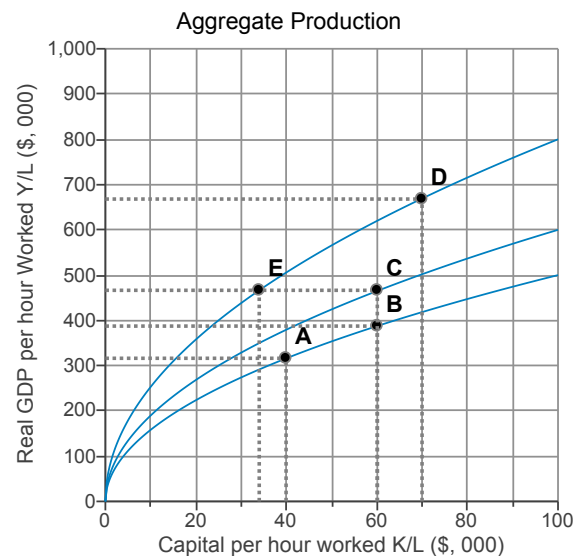
- ☐ A. they can save that money and invest in capital accumulation, which is much better for growth.
- ☐ B. knowledge capital is both rival and excludable, and no other firms can freely access the research and development of one particular firm.
- ☐ C. research and development are expensive and at the firm level, it is subject to increasing returns.
- ☒ D. knowledge capital is both nonrival and nonexcludable; other firms can freely access the research and development of one particular firm.

Government policy can increase the accumulation of knowledge capital in all the following ways except by:

- ☐ A. subsidizing research and development.
- ☐ B. protecting intellectual property with patents and copyrights.
- ☒ C. investing in capital accumulation.
- ☐ D. subsidizing education.

6. Refer to the diagram to the right, to answer the following:

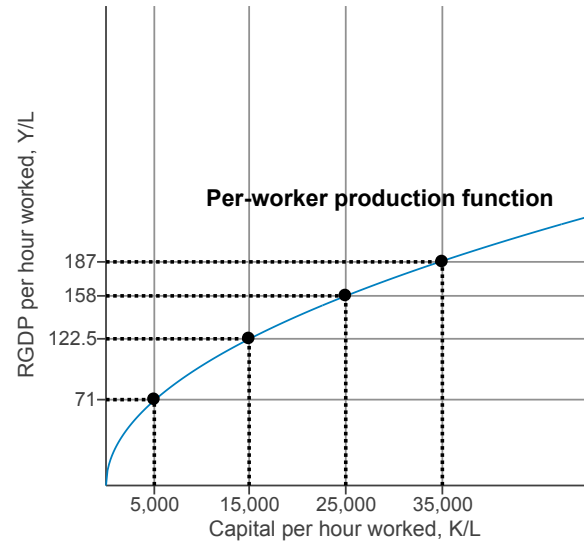
- a. The movement from point **A** to point **B** represents more capital per hour worked .
- b. The movement from point **B** to point **C** represents technological change .
- c. The movement from point **A** to point **C** represents technological change and more capital per hour worked .



7. The economic growth model explains growth in real GDP per capita in the long run. Because of the importance of labor productivity in explaining economic growth, the economic growth model focuses on the causes of increases in long-run labor productivity. What are the key factors that determine labor productivity? (Mark all that apply.)

- ☒ A. Quantity of capital per hour worked
☐ B. Efficiency wages
☐ C. Trade
☒ D. Technological change

8. Consider the per-worker production function. Equal increases in the quantity of capital per hour worked lead to diminishing increases in output per hour worked.



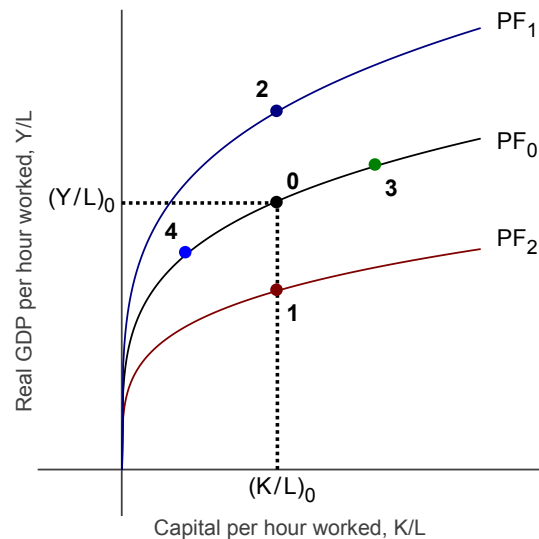
9. The graph below shows three production functions.

PF₀ shows the current production function for the economy, and capital per worker is currently (K/L)₀; thus output per worker is (Y/L)₀.

Suppose that there is a reduction in the level of technology.

Which of the points on the graph could show the new capital per worker and output per worker?

- ☐ A. 4 ☐ B. 2
☐ C. 3 ☒ D. 1



10. Technological change¹ is more important for economic growth than additional capital.

1: Definition

Technological change A change in the quantity of output a firm can produce using a given quantity of inputs.

11. An economy that does not experience increases in technological progress

- ☐ A. can sustain strong levels of economic growth as long as the economy continues to increase capital.
 - ☒ B. can experience economic growth by increasing capital, however, this will eventually stagnate and the economy will not continue to
 - ☐ C. will not be able to create economic growth unless the country has an increase in the labor force, either by population growth or immigration.
 - ☐ D. will not be able to experience any economic growth even with increases in capital.
-

12. Globalization entails all of the following *except*:

- ☐ A. Foreign portfolio investment.
- ☐ B. Foreign direct investment.
- ☒ C. Cultural exchange between nations.
- ☐ D. Open trade and technology transfers between countries.

Developing countries have benefited from globalization, because globalization can do all of the following *except*

- ☐ A. help spread new and advanced technology and aid in capital accumulation in the developing world.
 - ☒ B. impose trade barriers and tariffs on imported goods so as to protect domestic industries.
 - ☐ C. take poor countries out of a low saving-investment trap by providing foreign direct investment and foreign portfolio investment.
 - ☐ D. open the developing countries' market to foreign trade and thus help create a more robust economy.
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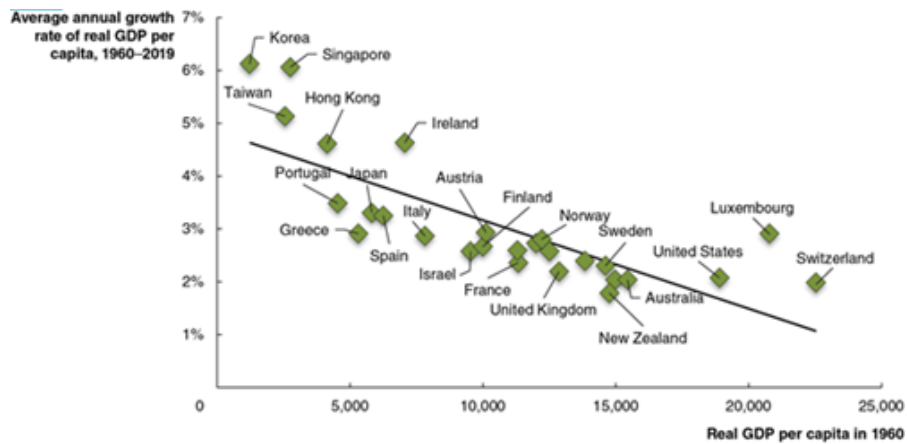
13. A study by the McKinsey Global Institute reported that labor productivity increased at an average annual rate of 5.8 percent between 1999 and 2013 in Mexico's large companies, but fell at an average annual rate of 6.5 percent over the same period for Mexico's smaller firms, such as family-owned stores and bakeries.

Source: Anthony Harrup, "Two Economies Explain Mexico's Productivity Quandary," *Wall Street Journal*, March 27, 2014.

Productivity growth would be much higher for Mexico's largest companies than for its smaller companies because the

- ☐ A. larger companies have more employees, so they are more likely to be included in statistical information.
 - ☐ B. smaller companies are more highly regulated and so have lower productivity growth.
 - ☒ C. larger companies have greater access to better technology which stimulates productivity growth.
 - ☐ D. smaller companies cannot achieve the scale to compete with larger companies.
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14.



The graph shows the relationship between country's initial level of GDP per capita and its growth rate over the following years. The horizontal axis shows the initial level of GDP per capita, and the vertical axis shows the rate at which GDP per capita is growing. Use the graph to help determine which of the following statements regarding economic growth are true:

- ☐ A. High-income countries should be in the upper-right part of the graph.
- ☒ B. Korea, Taiwan, and Singapore had the lowest incomes in 1960 and grew the fastest between 1960 and 2019.
- ☐ C. Low-income countries should be in the lower-left part of the graph.
- ☐ D. Countries like Switzerland and the United States that had the highest incomes in 1960 grew the fastest.

15. Suppose you are discussing global trade with a friend who insists a country would be better off by restricting trade and investment with other countries.

Which of the following economic responses would be the most logical for your discussion?

- ☒ A. I am not sure I agree. Countries that allow more globalization have experienced higher rates of economic growth and typically can utilize greater levels of foreign direct investment to increase economic growth.
- ☐ B. I am in agreement with you. Global trade doesn't seem to matter to growth rates and countries have greater economic growth when their growth comes from domestic investment.
- ☐ C. I am in agreement with you. Most low income countries allow high levels of trade with other countries. They experience high levels of trade but do not experience high rates of economic growth.
- ☐ D. I am not sure I agree. Low income countries usually have high levels of foreign direct investment which means they will eventually experience high levels of economic growth.

16. Which of the following is a factor explaining why low-income countries have not experienced rapid economic growth?

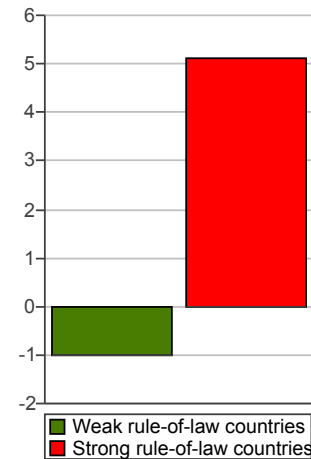
- ☐ A. Low-income countries have governments that prefer more simple standards of living.
- ☐ B. Low-income countries are generally expected to grow more slowly than high-income countries.
- ☒ C. Many low-income countries have poor public education and health.
- ☐ D. Low-income countries have started to grow too late and can never catch up to rich countries.

17.

The figure to the right illustrates the relationship between weak and strong rule-of-law² countries and economic growth. In addition to a country's failure to enforce rule-of-law, what else explains why more low-income countries do NOT experience rapid growth as the catch-up line predicts?

- ☐ A. Inability to borrow money needed for investment
- ☐ B. Shortage of childhood vaccinations
- ☐ C. Lengthy civil wars
- ☒ D. All of the above

Average annual growth rate real GDP per capita during the 1990s (%)



2: Definition

Rule of law The ability of a government to enforce the laws of the country, particularly with respect to protecting private property and enforcing contracts.

18. Two students, Ryan Wattenberg and Emma Bennett, are discussing the idea of convergence over coffee. Ryan considers convergence to be true in theory but impractical in the real world. He claims that most low-income developing countries are stuck in a cycle of poverty and so cannot catch up with developed countries. With increased globalization, Emma feels that the developing countries are growing and will converge with higher-income countries eventually. Zoey Smith, a friend of theirs, however thinks that the evidence on convergence is rather unclear. Despite the fact that developing countries are growing much faster than the developed countries, she thinks that they will not be able to catch up with the developed nations in the near future.

Which of the following, if true, will strengthen Emma's argument that developing countries will converge with developed countries?

- ☐ A. Most developing nations are at the point where the cost of capital exceeds the marginal product of capital.
- ☒ B. The number of bilateral and multilateral agreements relating to technology transfer from developed to developing nations has significantly increased.
- ☐ C. Since cash flows from foreign sources are considered volatile, many developing countries have capital controls and other regulations that restrict the flow of foreign investment.
- ☐ D. With low public investment in healthcare, the cost of private healthcare as a proportion of average income is very high in developing countries.
- ☐ E. Unlike the developed nations, most developing countries do not have extensive government-sponsored social security programs.

19. Two students, Ryan Wattenberg and Emma Bennett, are discussing the idea of convergence over coffee. Ryan considers convergence to be true in theory but impractical in the real world. He claims that most low-income developing countries are stuck in a cycle of poverty and so cannot catch up with developed countries. With increased globalization, Emma feels that the developing countries are growing and will converge with higher-income countries eventually. Zoey Smith, a friend of theirs, however thinks that the evidence on convergence is rather unclear. Despite the fact that developing countries are growing much faster than the developed countries, she thinks that they will not be able to catch up with the developed nations in the near future.

Which of the following, if true, will support Zoey's view that developing countries will not be able to catch up in the near future in spite of higher growth?

- ☐ A. A recent financial crisis adversely affected liquidity and credit in developed nations while causing a slump in developing countries' exports.
- ☐ B. The total land area covered by developing countries is much larger than the area covered by developed countries.
- ☐ C. Many developing countries are now opening up their economies to foreign investment
- ☐ D. Due to lower labor costs for skilled workers, many firms in developed nations outsource their business processes to developing countries.
- ☒ E. The difference in the current levels of real GDP between these two groups of countries is very high.

20. Economics arrives at the conclusion that economic growth will always improve economic well-being. Do you agree?

- ☒ A. Yes, economic growth increases living standards, improves health and education, and builds a corruption-free society.
 - ☐ B. No, economic growth has caused global warming that creates a very dangerous world for this and future generations.
 - ☐ C. No, economic growth increases pollution, depletes natural resources, and creates a new kind of health problem.
 - ☐ D. No, economic growth has empowered multinational companies who often exploit workers and the environment of the host country.
-

21. Healthier, more educated workers tend to be more productive. Greater overall productivity per hour worked is a fundamental component of long-term economic growth. However, many very successful individuals often find few opportunities in their own developing countries, and leave them for industrial countries.

By improving health and education, developing countries can generate economic growth, and increase incomes. This will help combat the prevalence of educated people leaving their home countries for opportunities elsewhere. That is, it will combat

- ☒ A. the brain drain.
 - ☐ B. corruption.
 - ☐ C. political reform.
 - ☐ D. communism.
-

22. Which of the following is a way in which governments can increase productivity through health and education?

- ☒ A. Governments can promote vaccinations for children and improve access to clean water.
 - ☐ B. Governments can strengthen property rights and enforce laws.
 - ☐ C. Governments can limit foreign competition by imposing tariffs and quotas on trade.
 - ☐ D. Governments can create tax incentives for both businesses and individuals.
-

23. Strong rule-of-law countries grow more rapidly than weak rule-of-law countries. What factor will most likely improve economic growth in weak rule-of-law countries?

- ☐ A. Communism
 - ☐ B. Capitalism
 - ☐ C. Corruption
 - ☒ D. Political reform
-

24. Which of the following is a way in which governments can directly promote savings and investment?

- ☐ A. Governments can promote vaccinations for children and improve access to clean water.
 - ☒ B. Governments can create tax incentives for both businesses and individuals.
 - ☐ C. Governments can expand primary and secondary schools.
 - ☐ D. Governments can limit foreign competition by imposing tariffs and quotas on trade.
-

25. Property rights have been determined to be a major factor that helps countries sustain economic growth. Governments can change laws and policies in order to give individuals and firms more freedom.

By doing so, government can help promote growth because greater property rights

- ☒ A. give entrepreneurs more incentive to take risks that create new products, ideas, and more technology.
 - ☐ B. create more incentives for investors to spend money on consumption goods that lead to increased levels of demand.
 - ☐ C. limit the interactions of workers across industries which leads to greater economies of scale.
 - ☐ D. encourage individuals to become more concerned with the health options for everyone in the general population.
-

26. The country Panjim has been growing at the rate of 8 percent annually following a series of economic reforms. Adelphia, a neighboring country, is also growing rapidly, but at a rate that is slightly lower than Panjim's. Elaine Mack and Prisca Baresi, who live in Adelphia, are discussing whether Adelphia's economy will surpass Panjim soon. Elaine is of the opinion that the high level of sales of capital goods indicates that growth in Adelphia will only increase further. Prisca however reminds Elaine that the working population in Adelphia is shrinking, which will actually reduce its growth prospects.

Which of the following, if true, will weaken Elaine's view that growth in Adelphia will surpass growth in Panjim?

- ☐ A. The capacity utilization rate in Adelphia has improved substantially in the recent past.
 - ☒ B. Adelphia has a larger population than Panjim but the level of GDP in Panjim is twice as high as that of Adelphia.
 - ☐ C. In the past couple of years, most of the science and technology patents have been granted to foreign-born individuals who live and work in Adelphia.
 - ☐ D. Adelphia's manufacturing exports to Panjim have increased sharply.
 - ☐ E. The level of public investment in health care has increased in Adelphia much more than it has increased in Panjim.
-

27. What is the difference between foreign portfolio investment and foreign direct investment?

When a firm builds or purchases a facility in a foreign country, foreign direct investment occurs.

When an individual or a firm buys stocks or bonds issued in another country, foreign portfolio investment occurs.

Briefly explain how a poor country might benefit from foreign portfolio investment or foreign direct investment.

- ☐ A. It can enable poor countries to become more independent and therefore less dependent on the global economy
 - ☒ B. It can give low-income countries access to technology and funds that otherwise would not be available.
 - ☐ C. The gains are often captured by the powerful elite to be equitably distributed to households in the bottom quintiles.
 - ☐ D. Households will be able to save less and spend more, thus providing firms with the funds necessary for investment.
-

28. For each of the following policies, indicate whether it will or will not increase the rate of economic growth in Canada.

- a. The government passes an investment tax credit, which reduces a firm's taxes if it installs new machinery and equipment. It is likely that the rate of economic growth will increase .
- b. The government shortens the period of patent protection for new inventions. It is likely that the rate of economic growth will decrease .
- c. The government provides more funds for low-interest loans to college and university students. It is likely that the rate of economic growth will increase .